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Use of nanomaterials in e-nose applications is gaining ground, and so is the capability to create sensors with ultra-high sensitivities and fast response (partly due to a smaller structure). Smaller sensor size allows integration into a larger number of devices. An attractive class of materials for functional nanodevices is metal-oxide semiconductors. These offer the advantages of simple operation, ease of fabrication and compatibility with microelectronic processing, as well as low cost and low power consumption.

As regards e-tongue, within a few years, researchers anticipate a machine that determines the precise chemical structure of food and why people like it. Digital taste buds will also help us to eat smarter and healthier.

E-tongues are used in liquid environments to classify contents of the liquid, identify the liquid itself or discriminate between samples. Most e-tongues are based either on potentiometric or amperometric sensors. Taste sensors have artificial polyvinyl chloride (PVC)/lipid membranes that interact with a target solution such as caffeinated beverages. Change in the potential of the lipid membrane is the sensor output or measurement. Investigating potential change results in measuring the taste provided by the output of the chemical substances. Multiple sensors of the sensor array provide this output to form a unique fingerprint.

Hearing systems are increasingly being trained by listening to sounds, detecting patterns and building models to decompose sounds. One of the most common applications for sensors in this segment is hearing aids. Digital advances have made today's hearing aids smaller, smarter and easier to use. The most advanced hearing aids can interact with other devices, such as smartphones and digital music players, to deliver sounds directly and wirelessly to the listener.

Recent improvements are based on better microprocessors and noise-reduction software so that the hearing aid can be selective about the types of sound it amplifies, muffles or suppresses. Much of current research is focused on directionality and speech enhancement. Sound systems can employ digital-signal processing to

automatically shift between two different types of microphones in order to pick up either a single speaker's voice or sound coming from all around.

Digital-speech enhancement can now increase the intensity and audibility of some segments of human speech. Research projects are underway to reduce the size and cost of hearing aids, improve their directional capabilities, and identify and amplify desired sounds such as a human voice while muting background noise.

Electronic-skin prototypes are stretchy, thin films that can sense temperature, pressure and even monitor blood oxygen or alcohol levels. But most of these devices are missing hair—a key feature of real skin that allows us to feel a wider range of conditions.

Now researchers have combined hair-like wires with electronic skin to make a more versatile sensor for robots, prosthetics and other applications. Robots and prosthetics are becoming ever more human-like, but the electronic skins designed to enhance their usefulness don't yet have the full range of tactile senses that we have. To capture that sensation, some researchers have developed separate sensors that mimic the fine hair by sensing and detecting air flow.

Looking ahead

Thanks to significant advances in micro-fluidics and electronics, e-nose and e-tongue technologies will evolve both in terms of robustness and reduction in the current needed to optimise each application—a process that requires significant investment of time and resources. Miniaturisation will further extend flexibility.

Beyond food, e-sense technology may find applications in environmental analysis to detect water contamination or illicit drugs; in clinical diagnostics to monitor saliva, sweat or urine; and in agriculture to detect fungal contamination in feed. However, both e-nose and e-tongue technologies need improvements in sampling procedures (by reducing clean-up or extraction before analysis, for example) and reduction of carry-over/environmental noise (for instance, in the form of moisture contamination), which affect sensor drift and sensitivity. **EFY**